

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\bar{\Phi} = c_0 \left(\sin \frac{\vartheta}{2} \right)^m \left(\cos \frac{\vartheta}{2} \right)^m e^{i(m\varphi - Et)} \begin{pmatrix} c(\cos \frac{\vartheta}{2}) P_n^{(m-1/2, m+1/2)} A(r) \\ c(\sin \frac{\vartheta}{2}) P_n^{(m+1/2, m-1/2)} A(r) \\ (\cos \frac{\vartheta}{2}) P_n^{(m-1/2, m+1/2)} B(r) \\ -(\sin \frac{\vartheta}{2}) P_n^{(m+1/2, m-1/2)} B(r) \end{pmatrix}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} |I_{\omega, \mathbf{c}}(U(t)) - I_{\omega, \mathbf{c}}(\Phi)| &\leq 2|L(U(t)) - L(\Phi)| + \frac{5}{2}|N(U(t)) - N(\Phi)| + |\mathbf{c} \cdot (\mathbf{P}(U(t)) - \mathbf{P}(\Phi))| \\ &\leq C(1 + \|U(t)\|_{\mathcal{H}^1}^2 + \|\Phi\|_{\mathcal{H}^1}^2) \|U(t) - \Phi\|_{\mathcal{H}^1} \\ &\leq C(1 + M^2 + \mu_{\omega, \mathbf{c}}) \|U(t) - \Phi\|_{\mathcal{H}^1} \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathcal{N}_2(h, \omega) &= -[|\partial_x|, \partial_x T_{\partial_x}]h - iT_{\partial_x^2}|\partial_x|^{1/2}\omega + i[|\partial_x|^{1/2}, T_{\partial_x}\partial_x]\omega \\ &\quad + |\partial_x|M_2(\omega, h) - i|\partial_x|^{1/2}Q_2(\omega, \omega) - i|\partial_x|^{1/2}T_{|\partial_x|^3h}P_{\geq 1}h. \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathcal{L}_K &= \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - (1 - g^2\zeta\eta^2)\tilde{G}_\mu\tilde{G}^\mu + 2(m + gh\zeta\eta^2)A_\mu\tilde{G}^\mu + \\ &\quad - 2h\zeta\eta^2\Sigma\partial_\mu A^\mu + h^2\zeta\eta^2A_\mu A^\mu + \zeta\eta^2\partial_\mu\Sigma\partial^\mu\Sigma, \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \int_{A_1 \cap A_2} |g - g_\rho| \, d\pi &\leq \int_{A_1 \cap A_2} M \left[\mathbf{d}_{S_1}(s_1^*, s_1)^{p'_1} + \mathbf{d}_{S_2}(s_2, s_1^*)^{p'_2} \right] \, d\pi(s_1, s_2) \\ &\leq B(\rho^{p'_1 - p_1} + \rho^{p'_2 - p_2}). \end{aligned}$$